



UY/CH-CHILD – A Public Chinese L2 Speech Database of Uyghur Children

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Abstract

Exploring the progression of pronunciation skills in second language (L2) acquisition among children presents an intriguing research avenue. Yet, the comprehension of this process for Uyghur children learning Chinese as their L2 has been constrained by a scarcity of speech data. To bridge this gap, we have developed the UY/CH-CHILD speech database, comprising 29,061 samples of Chinese words articulated by 106 Uyghur children from both kindergartens and primary schools. The database includes carefully labelled syllables and tones by native Chinese speakers. To showcase the utility of this novel resource, we conducted a comparative analysis of pronunciation errors between the kindergarten and primary school groups, unveiling interesting insights into the evolution of pronunciation proficiency in Uyghur children as they mature. The database can be downloaded online at <http://child.csit.org>.

Index Terms: child speech database, second language acquisition, Uyghur-Chinese

1. Introduction

Extensive research has delved into the intricacies of second language (L2) acquisition [1], particularly focusing on children [2, 3]. These investigations not only illuminate the psychological mechanisms underlying language acquisition but also offer valuable insights for crafting tailored educational programs and materials [4].

This study centres on Uyghur children learning Mandarin Chinese as their L2. Our specific emphasis lies in the development of pronunciation skills from kindergarten through primary school. The study holds practical significance, given the sizable Uyghur population in northwest China, where fluency and accuracy in Chinese can open up enhanced opportunities in life and career prospects, especially for the younger generation. Notably, the Uyghur-Chinese cross-lingual learning presents a unique case of ‘advanced L2 learning’ as discussed in [5], where children increasingly lean towards Chinese (L2) influence starting from kindergarten, sometimes surpassing their native Uyghur language. Numerous research avenues remain unexplored for this advanced L2 learning, such as investigating how the initial L1 environment influences L2 pronunciation in the long run, determining the optimal age to commence L2 education, and exploring whether pronunciation accuracy naturally improves with age without deliberate phonological monitoring and correction.

Conducting such research necessitates a robust database that meets specific criteria: (1) substantial in size; (2) cover-

ing the entire pronunciation acquisition process; and (3) featuring reliable annotations. Regrettably, existing databases fail to meet these criteria. The closest relevant resource may be the lexicon of pronunciation errors of Uyghur children aged 3-6 when producing Chinese words, as referenced in [6]. However, this dataset lacks speech data and remains inaccessible to the public.

To advance this research direction, this paper introduces UY/CH-CHILD, an open-access L2 Chinese speech database compiled from recordings of Uyghur children. The database is detailed as follows:

- It emphasizes pronunciation over vocabulary usage. Hence, children were instructed to articulate simple, standalone words, with precise pronunciations annotated at the syllable level in *Pinyin* by native speakers.
- It is extensive in scale, comprising 29,061 speech samples from 106 Uyghur children, enabling statistically significant conclusions to be drawn for various tasks like acoustic pattern analysis and assessing pronunciation biases.
- It encompasses children aged 4-8, segmented into kindergarten and primary school groups. This structure facilitates a comprehensive study of the entire process of pronunciation acquisition among Uyghur children, who typically commence learning Chinese in kindergartens and refine their pronunciation skills in primary schools.
- It is openly accessible and free.

It is underscored that UY/CH-CHILD serves as a versatile data resource applicable across multiple research domains. Firstly, it proves invaluable for investigating the acquisition of Chinese pronunciation as an L2 in a broader context, extending beyond research solely focused on Uyghur children. Notably, there is a scarcity of public databases targeting Chinese as an L2, especially for children learners. Secondly, owing to the distinct characteristics of different languages, UY/CH-CHILD can serve as supplementary material for exploring both commonalities and unique features in children learning various L2 languages. Thirdly, conducting comparative studies between Uyghur and Chinese children learning Chinese using this dataset can shed light on the influence of familial environments [7, 8, 9, 10, 11, 12, 13]. Lastly, the data can be utilized for tasks such as mispronunciation detection and attribution or other engineering studies.

The construction of this new database will be detailed in Section 3 and Section 4 will present a pronunciation bias analysis using this database. Section 5 will outline the conclusions and offer insights into the potential future.

This work was supported by the National Social Science Foundation of China (NSSFC) under Grants No.22XYY048.

2. Related Work

CHILDES stands out as a prominent project offering extensive data resources for studying children’s language development¹, with all materials freely accessible. However, databases in Chinese are scarce in the CHILDES repository. One notable exception is the English-Mandarin Singapore Corpus developed by Yow and colleagues², originally crafted to explore English and Chinese code-switching in bilingual preschoolers [14, 15]. Despite its value, this database lacks phonetic labels, making it unsuitable for assessing pronunciation skills. Another relevant database within CHILDES is the Xinjiang Children corpus³ curated by Prof. Zhou, featuring speech recordings of 15 children in Xinjiang aged 4-8. However, this corpus lacks phonetic labels in transcriptions and the limited number of children hampers robust statistical analyses.

The CHILDES repository also includes L1 Chinese databases spoken by children, as listed on the webpage <https://childes.talkbank.org/access/Chinese/>. While not directly applicable to L2 learning research, these resources can be leveraged for comparative studies. Noteworthy databases in this category include the Dinner-talk corpus⁴ and the Mandarin Zhou Narratives corpus⁵, along with the Mandarin Chang Personal Narrative Corpus by Zhou and Li [16, 17], the Mandarin Chang Personal Narrative Corpus⁶ by Chang et al. [18], and the Mandarin Beijing Corpus developed by Twila Tardif⁷ [19, 20]. Additionally, the longitudinal single speaker Tong corpus⁸ by Deng et al. [21] contributes to this collection.

Several private data sources exist for children speaking Chinese that are not publicly available but can be obtained by request from the authors. Notable examples include the CASS-CHILD database developed by Gao et al. [22] and the SingaKids-Mandarin database published by Chen et al. [5]. Limited accessibility restricts the use of these private databases. Furthermore, there are known Chinese L2 databases [23, 24, 25, 26], but as they feature adult speakers, their relevance to our research objectives is diminished. Table 1 provides a summary of the characteristics of the aforementioned databases.

Table 1: *Some known Chinese speech databases of children.*

	#Spk	Age	L1/L2/B	Phonetic Label	Open Source
Yow [14, 15]	55	5-6	B		✓
Zhou Xinjiang	15	4-8	L2		✓
Zhou [16]	72	5-6	L1		✓
Li [17]	200	3-6	L1		✓
Chang [18]	181	3-8	L1		✓
Tong [21]	1	1;7-3;4	L1		✓
Tardif [19]	11	1-2	L1		F
Gao [22]	23	1-4	L1	✓	
Chen [5]	255	7-12	B	✓	
UY/CH-CHILD	106	4-8	L2	✓	✓

Note: ‘Phonetic Label’ means true pronunciation labels in *Pinyin*. ‘L1/L2/B’ means ‘native language/second language/bi-lingual’. ‘F’ in ‘Open Source’ means access failed.

¹<https://childes.talkbank.org/>

²<https://childes.talkbank.org/access/Biling/Singapore.html>

³<https://childes.talkbank.org/access/Chinese/Mandarin/Xinjiang.html>

⁴<https://childes.talkbank.org/access/Chinese/Mandarin/ZhouDinner.html>

⁵<https://childes.talkbank.org/access/Chinese/Mandarin/ZhouNarratives.html>

⁶<https://childes.talkbank.org/access/Chinese/Mandarin/ChangPN.html>

⁷<https://childes.talkbank.org/access/Chinese/Mandarin/Beijing.html>

⁸<https://childes.talkbank.org/access/Chinese/Mandarin/Tong.html>

3. Database Construction

UY/CH-CHILD comprises a compilation of pronunciation samples for a predefined set of *target words*, with each sample annotated with the canonical pronunciation and the actual pronunciation in Pinyin format. The database construction process involves four key steps: (1) Vocabulary Design: selecting representative target words for Chinese pronunciation; (2) Speech Recording: children are guided to pronounce the target words under the supervision of a tutor; (3) Target Word Segmentation: identifying speech segments corresponding to clearly pronounced target words; (4) Phonetic Annotation: labelling the true pronunciation (in Pinyin) of the identified target words.

3.1. Vocabulary Design

The selection of target words was based on the Articulation Test for Mandarin Chinese-speaking Preschoolers developed by the Institute of Linguistics at the Chinese Academy of Social Sciences (CASS) [22, 27]. This word list comprises 174 words consisting of 1-3 Chinese characters (syllables), encompassing key phonological features of Mandarin Chinese, including syllables, tones, syllable combinations, weak stress, and rhotacism. All words represent common everyday objects like eggs, sofas, flowers, hands, etc., facilitating easy comprehension for children. This approach ensures that the collected data primarily focuses on pronunciation accuracy rather than word usage proficiency.

From the full list of 174 words, two test sheets were created: Test A contains 138 simpler and easily recognizable words such as ‘hand’ and ‘flower’, while Test B includes 140 more challenging words like ‘crocodile’, ‘axe’, and ‘chalk’. There are 104 overlapping words between Test A and Test B. Test A was designated for recording sessions with younger children aged 4-5 in the kindergarten group, while Test B was used for older children in both the kindergarten and primary school groups.

3.2. Speech Recording

The speech data was recorded between 2022 and 2023 during two phases due to the COVID-19 pandemic: phase I in May 2022 and phase II in February 2023. All recordings took place in Ili Prefecture, Xinjiang Uyghur Autonomous Region in China. The participating children come from native Uyghur families and attend kindergartens or primary schools where Chinese is the primary language. Basic information about the children was collected, and parental or guardian consent for data usage was obtained under strict privacy protection regulations.

The speech data was captured in quiet rooms using a laptop, with recording parameters set at a 16kHz sampling rate, 16-bit precision, and single-channel configuration. The recording process closely adhered to the guidelines proposed in [28]. During each recording session, a designated tutor oversaw the procedure. Their role involved operating the laptop, displaying images of the target words to the children, and encouraging them to articulate the corresponding object names. For instance, if a picture of a sofa appeared on the screen and the child struggled to identify it, the tutor would prompt with a question like, ‘What do you like to sit on at home?’. If necessary, the tutor would play a prerecorded voice for the child to mimic. Corrections to pronunciation were not provided if the child had identified the correct word. Importantly, the entire session was recorded and saved to facilitate subsequent steps, including annotation and review.

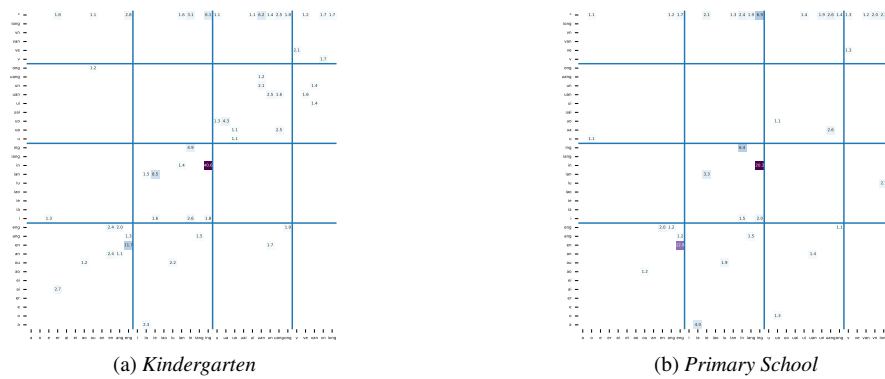


Figure 1: Confusion matrix of vowels for (a) the kindergarten group and (b) the primary school group. The x-axis denotes the target vowel and the y-axis indicates the pronounced vowel. Each element (x, y) displays the probability of vowel x being pronounced as vowel y. To enhance clarity, insignificant numbers (below 1.0%) and correct pronunciation probabilities are omitted from the representation.

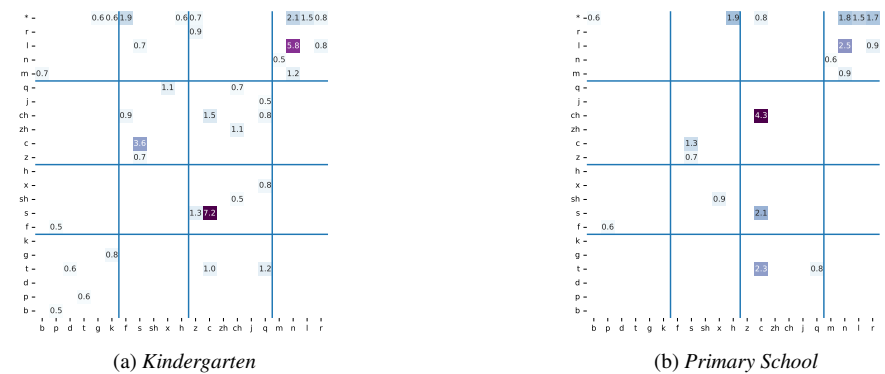


Figure 2: Confusion matrix of consonants for (a) the kindergarten group and (b) the primary school group. The notations are the same as in Figure 1. The insignificant numbers (below 0.5%) and correct pronunciation probabilities are omitted.

3.3. Target Speech Segmentation

All speech data from the recording sessions were uploaded to a web-based platform for target speech segmentation. Students from the International Cultural Exchange College at Xinjiang University were enlisted to perform this task. Their responsibility was to identify all speech segments clearly articulated by the children, regardless of content.

3.4. Phonetic Annotation

The same web-based platform was utilized for annotating the pronunciations of the segmented target speech. Students from Xinjiang University were engaged in this labelling process. They were tasked with evaluating the pronunciation accuracy of the target speech segments and annotating any deviations from the canonical pronunciation in Pinyin. Each Chinese character's Pinyin annotation comprised three components: the consonant part (Initial), the vowel part (Final), and the tone, structured as Initial~Final~Tone. In cases where classification posed challenges, the label '*' was used. For example, if the canonical pronunciation of the character '光' is 'guang1', a label 'g~uan~*' indicates that the final (vowel) 'uang' was mispronounced as 'uan', with uncertainty '*' regarding the tone.

4. Pronunciation Bias Analysis

To showcase the utility of UY/CH-CHILD, a pronunciation bias analysis was conducted for both the kindergarten and primary school groups. This analysis sheds light on the enhancement of Uyghur children's pronunciation skills as they progress through

Chinese education. Before delving into the analysis, a brief overview of the phonological characteristics of Chinese is provided.

4.1. Chinese Phonology

Chinese characters represent syllables, each typically comprising a consonant (Initial) and a vowel (Final), pronounced with one of four tones (T1-T4) denoting pitch height and contour. While some syllables may end with the optional lightening and addition of the suffix *er*, such complexities are not considered in UY/CH-CHILD.

For many Chinese learners, two primary challenges arise [29]: (1) Compound vowels, where a vowel consists of multiple connected vowels (e.g., *ao*) sometimes accompanied by a nasal element (e.g., *ang*), leading to subtle yet crucial pronunciation distinctions. (2) Tonal accuracy is vital, but producing clear and distinguishable tones can be challenging, particularly for individuals from non-tonal language backgrounds, such as Uyghur.

4.2. Vowel Bias Analysis

Figure 1 illustrates a *confusion matrix* of vowels, representing the probability that a vowel on the x-axis is pronounced as a vowel on the y-axis. The '*' label on the y-axis denotes unclear pronunciations. The vowels are categorized into 4 groups according to the speaking style: open mouth, even teeth, closed mouth, and rounded mouth. These groups are delineated by blue lines in Figure 1, with the blue squares along the diagonal representing 'confusion areas'. Notably, the primary school group exhibits fewer errors compared to the kindergarten group,

Table 2: Comparison between Uyghur Children and native Chinese Children.

		Vowels	Consonants	Tone
Commonality		- Acquisition improves with age - Lower error rate for monophthongs	- Acquisition improves with age - Affricates are difficult for children	No clear commonality
Specificity	Native	- More errors in deleting 'n', 'ng' - Replacing 'n' with 'ng' - Triphthongs often simplified to diphthongs or monophthongs	- Consonants j, q, x are rare in other languages - By the age of 4;6, most children can correctly use consonants	Tone errors are very rare even in the youngest children
	Uyghur	Interchanging of front and back nasals (e.g., en-eng, in-ing)	Slow acquisition of consonants, with a high error rate	Tone errors are common among learners of all groups

indicating improved pronunciation skills with increased education duration.

In terms of error types, rare errors are observed on individual vowels, even among kindergarten children. Most errors occur with the second vowel or the final nasal sound, such as *ua* → *uo*, *ing* → *in*. Errors involving final nasals are particularly significant; for the primary school group, many errors evident in the kindergarten group diminish, except for confusion between nasal/non-nasal vowels and the two distinct nasals *n/ng*.

Table 3 highlights the top 5 vowels with the highest error rates and significant error types. Once again, compound vowels, especially nasalized compound vowels, exhibit a higher error frequency [30, 31]. This trend aligns with the absence of compound vowels in Uyghur [32, 33].

Table 3: Vowels with top-5 error rates (ER%) and top-3 errors for each of the vowels.

Kindergarten				Primary School			
No.	Ph.	ER%	Errors	No.	Ph.	ER%	Errors
1	ing	49.4	in, *, i	1	ing	29.9	in, *, i
2	eng	17.8	en, *, ang	2	eng	16.4	en, *, ang
3	ie	11.6	ian, i, in	3	in	10.7	ing, *, i
4	uan	11.1	*, un, uang	4	un	5.6	*, en, u
5	in	10.8	ing, *, i	5	uang	5.5	ua, *, ong

4.3. Consonant Bias Analysis

The confusion matrix of consonants is depicted in Figure 2, while Table 4 outlines the consonants with top-5 error rates. The blue lines in Figure 2 mark four categories: fricatives, stops, affricatives, nasal and lateral. Notably, the frequency of pronunciation errors on consonants is notably lower compared to vowels. Furthermore, there is a clear reduction in errors within different consonant categories from kindergarten to primary school levels, suggesting an improvement in phonetic accuracy as children progress in their Chinese language education.

The majority of errors occur on consonants that are either absent in Uyghur or differ in pronunciation style from Uyghur consonants, such as *c*, *s*, and *r*. Confusion is also observed between nasal consonants like *m* and *n*, as well as between the nasal *n* and the lateral *l*. Even for native speakers, distinguishing these sounds can be challenging [34]. The most prevalent error pair identified is *c* → *ch*. This mispronunciation arises because both phonemes are absent in Uyghur.

4.4. Tone Bias Analysis

The analysis on tones is presented in Figure 3, illustrating the confusion matrix. Notably, the second and third tones exhibit the highest degree of confusion. This observation is expected, given that the third tone is the most intricate, involving an intermediate shift from a rising tone (similar to the second tone) to a falling tone (akin to the fourth tone) [35, 36].

Table 4: Consonants with top-5 error rates (ER%) and top-3 errors for each of the consonants.

Kindergarten				Primary School			
No.	Ph.	ER%	Errors	No.	Ph.	ER%	Errors
1	c	11.3	s, ch, t	1	c	9.8	ch, t, s
2	n	9.4	l, *, m	2	n	5.1	l, *, m
3	s	5.8	c, z, l	3	s	3.6	c, z, zh
4	q	3.9	t, x, ch	4	r	2.6	*, l, b
5	ch	3.2	zh, q, sh	5	h	2.5	*, p, k

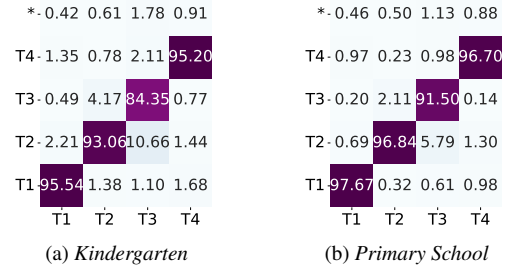


Figure 3: Confusion matrix of tones for (a) the kindergarten group and (b) the primary school group.

4.5. Comparison with native children

We contrast our experimental findings with the research outcomes concerning native Chinese children documented in the literature [37]. Table 2 provides a summary of this comparison, revealing that Uyghur children exhibit distinct characteristics in their Chinese language learning process, with the greatest challenge being tone acquisition. These insights offer valuable guidance for Chinese L2 education tailored to Uyghur children.

5. Conclusion

In this study, we have introduced a Chinese L2 speech database comprising recordings from 106 Uyghur children, meticulously annotated for pronunciation accuracy. The database serves as a valuable resource for investigating children's pronunciation acquisition. A preliminary analysis was conducted using this database to explore the development of pronunciation skills among Uyghur children at different educational stages. The findings indicate: (1) Pronunciation errors predominantly occur in vowels, particularly in compound vowels with nasal codas; (2) For errors on consonants, the most prevalent mistake is the mispronunciation of the phoneme *c* as *ch*; (3) Tones 2 and 3 pose the most confusion; (4) Notable enhancement in pronunciation skills is evident from the kindergarten to primary school groups. Due to space constraints, further experiment details will be elaborated in an extended journal publication.

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